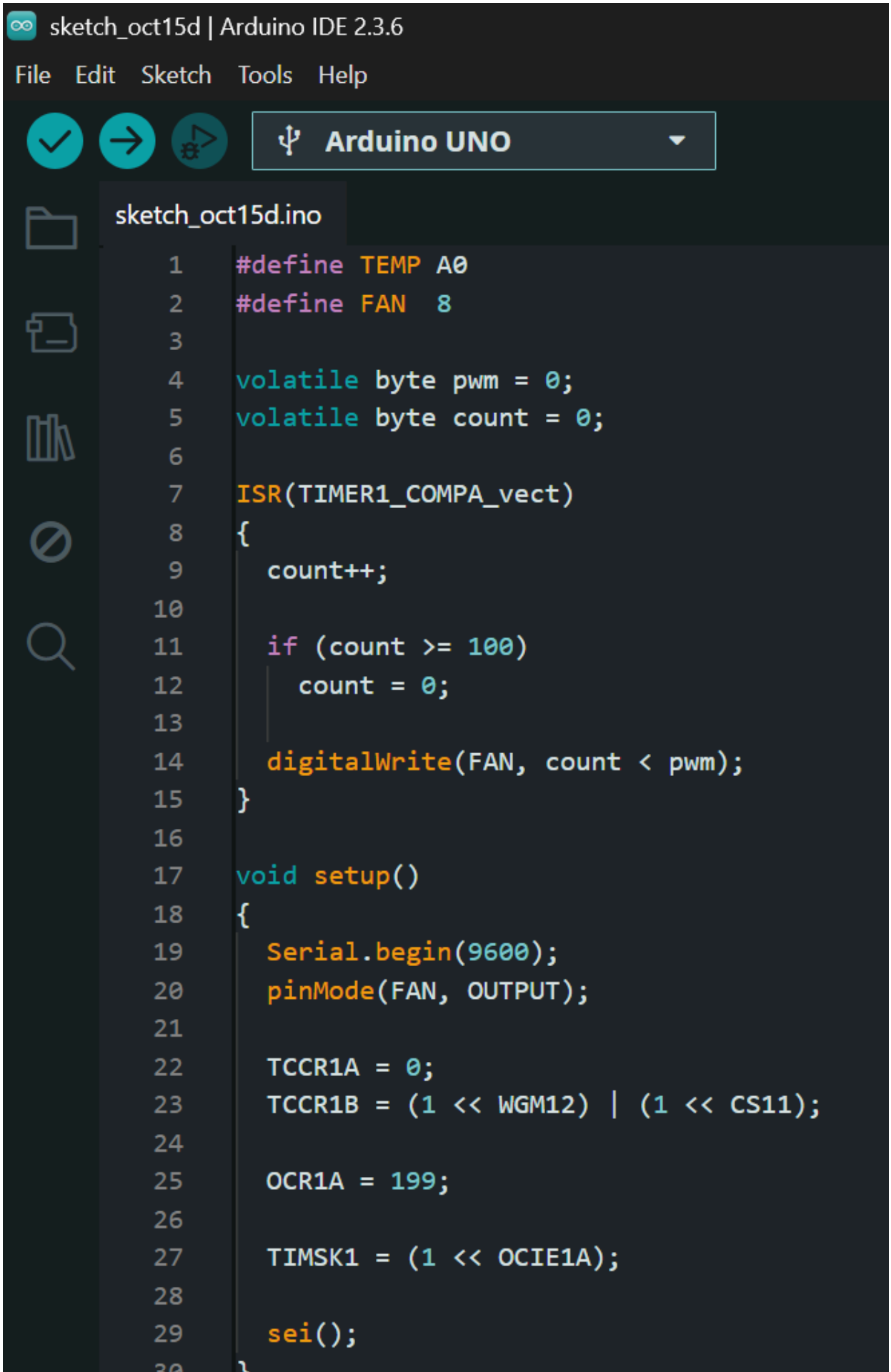
Arduino Code:


sketch_oct15d | Arduino IDE 2.3.6
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sketch_oct15d.ino
#define TEMP A0
#define FAN 8
volatile byte pwm = 0;
volatile byte count = 0;
ISR(TIMER1_COMPA_vect)
{
count++;
if (count >= 100)
count = 0;
digitalWrite(FAN, count < pwm);
}
void setup()
{
Serial.begin(9600);
pinMode(FAN, OUTPUT);
TCCR1A = 0;
TCCR1B = (1 << WGM12) | (1 << CS11);
OCR1A = 199;
TIMSK1 = (1 << OCIE1A);
sei();

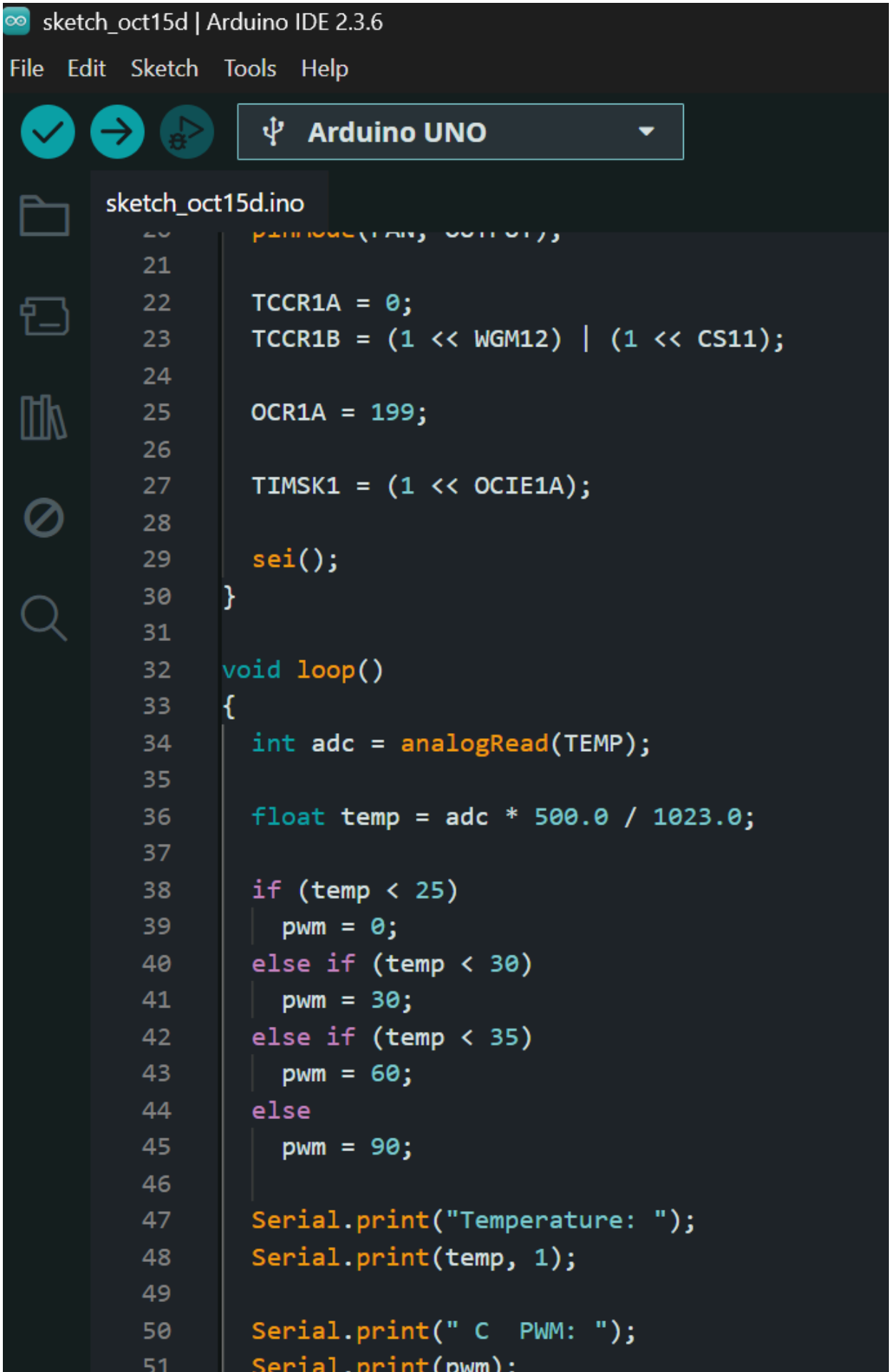
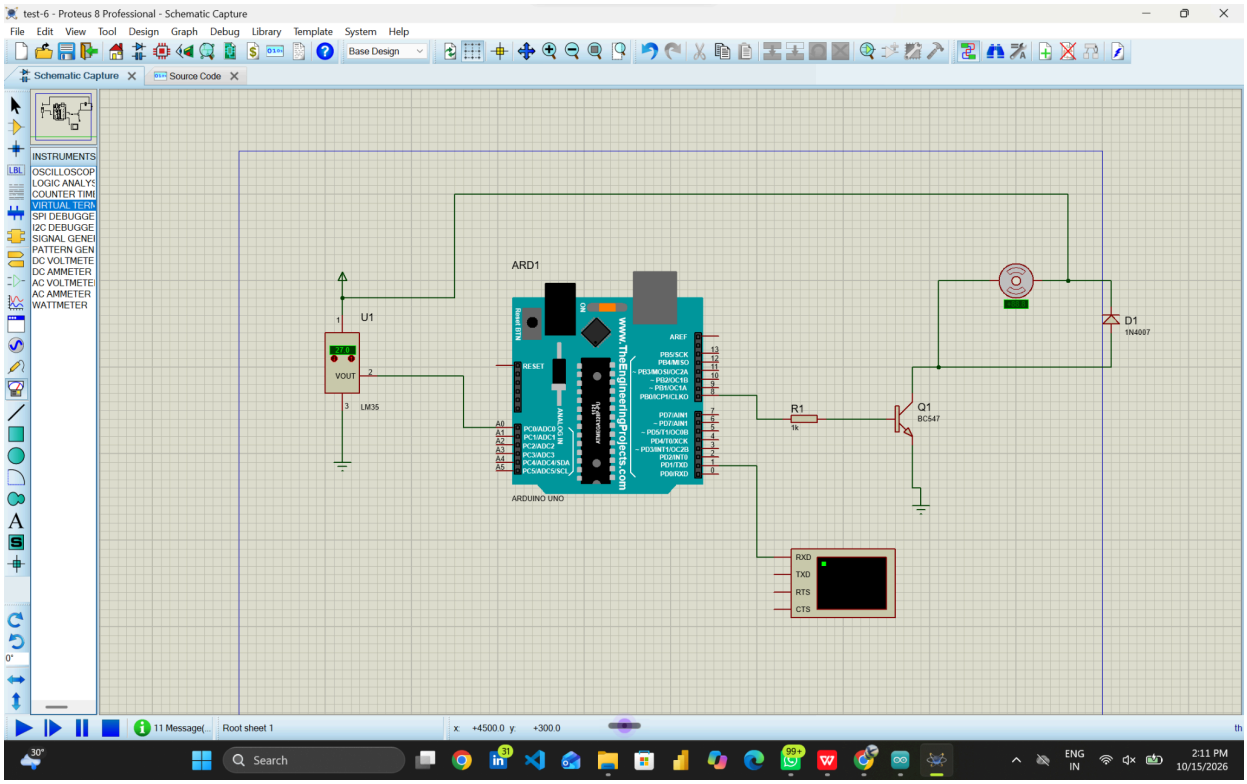

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21
22 TCCR1A = 0;
23 TCCR1B = (1 << WGM12) | (1 << CS11);
24
25 OCR1A = 199;
26
27 TIMSK1 = (1 << OCIE1A);
28
29 sei();
30 }
31
32 void loop()
33 {
34 int adc = analogRead(TEMP);
35
36 float temp = adc * 500.0 / 1023.0;
37
38 if (temp < 25)
39 pwm = 0;
40 else if (temp < 30)
41 pwm = 30;
42 else if (temp < 35)
43 pwm = 60;
44 else
45 pwm = 90;
46
47 Serial.print("Temperature: ");
48 Serial.print(temp, 1);
49
50 Serial.print(" C PWM: ");

Diagram:



Output:

